

Haryana LEET / OCET 2026

Section D — Other Engineering Streams

20 Questions | 20 Marks | No Negative Marking

d-1 CIVIL ENGINEERING COURSES (5 Q | 5 M)

■ Bricks & Building Materials

- Raw materials for bricks: clay, sand, lime, iron oxide, magnesia
- Manufacturing: preparation → moulding → drying → burning (clamp/kiln)
- Good brick: regular shape, uniform colour, 3-4 kg weight, rings when struck
- Types of cement: OPC, PPC (pozzolana), RHC (rapid hardening), SRC (sulphate resistant)
- Brick bonds: English bond (strongest), Flemish bond (aesthetic), stretcher bond

■ Soil Mechanics

- Index properties: void ratio $e = V_v/V_s$ | porosity $n = V_v/V$
- Degree of saturation $S = V_w/V_v$ | water content $w = W_w/W_s$
- Seepage: flow of water through soil | Darcy's law $v = ki$ (k = permeability)

■ Water Supply Engineering

- Per capita demand: 135 lpcd (domestic) — IS standard
- Sources: surface (river, lake) & groundwater (well, tube well, spring)
- Treatment steps: aeration → sedimentation → coagulation → filtration → disinfection
- Sludge treatment: thickening → digestion → drying | Disposal: land, incineration

■ Foundation & RCC

- Shallow foundations: strip/wall, spread/isolated, combined, raft/mat
- Deep foundations: pile (friction/bearing), well/caisson
- Concrete grades: M10, M15, M20 (general), M25, M30 (structural)
- W/C ratio: lower → higher strength | Compaction: removes air voids
- Workability tests: slump test, compaction factor test, Vee-Bee test

■ Surveying Basics

- Chain surveying: direct measurement, triangulation network
- Bench Mark (BM): reference point for levels — RL of BM known
- Compass surveying: WCB (whole circle bearing 0° - 360°) | QB (quadrant bearing)
- Fore bearing & back bearing: differ by 180°

■ Roads, Bridges & Irrigation

- Road materials: WBM (water bound macadam), bitumen, cement concrete
- Flexible pavement: bituminous, distributes load gradually
- Rigid pavement: cement concrete, distributes load over wider area
- Bridge classification: by material (timber/steel/RCC), by span, by use
- Methods of irrigation: surface (furrow, border, check), sprinkler, drip
- Dams: gravity, arch, earth, rock-fill | Site: narrow gorge with strong rock

■ Doors, Roofs & Ecology

- Types of doors: panel, flush, glazed, louvred, collapsible, rolling shutter
- Types of roofs: flat (RCC slab), pitched (sloped, tiles), shell, dome
- Classification of rocks: igneous (granite), sedimentary (sandstone), metamorphic (marble)
- Water pollution: industrial effluents, agricultural runoff, sewage
- Non-conventional energy: solar PV, wind turbine, biogas, small hydro

d-2 TEXTILE ENGINEERING COURSES (5 Q | 5 M)

■ Fibres

- Natural fibres: plant-based (cotton, jute, flax) | animal-based (wool, silk)
- Synthetic fibres: nylon (polyamide), polyester, acrylic, spandex
- Properties: length (staple vs filament), fineness, strength, elongation

■ Yarn & Fabric Manufacturing

- Spinning process: opening → carding → drawing → roving → spinning
- Ring spinning (traditional) vs Open-end (rotor) spinning (modern, faster)
- Air-jet spinning: uses compressed air — fastest modern method
- Weaving: interlacing warp & weft yarns on a loom
- Basic weaves: plain, twill, satin | Fabric: woven, knitted, non-woven

■ Textile Testing & Quality Control

- Yarn count: indirect (Ne — more count = finer) | Direct (Tex, Denier — more = coarser)
- Tensile strength, elongation, twist per inch (TPI)
- Fabric: GSM (g/m²), thread count, pilling resistance

■ Bleaching, Dyeing & Printing

- Bleaching: removes natural colour using H₂O₂ (cotton), Na₂S₂O₄ (synthetic)
- Dyeing: direct dyes, reactive dyes, disperse dyes (for polyester)
- Printing: screen printing (flat/rotary), digital (inkjet), block printing

d-3 CHEMICAL / PRINTING TECHNOLOGY (2 Q | 2 M)

■ Chemical Engineering Basics

- Fluid flow: laminar vs turbulent | Bernoulli's equation application in pipes
- Chemical process industries: fertilizers (Haber process), sulphuric acid (contact process)
- Agro-based industries: sugar, paper, alcohol, vegetable oil
- Petrochemicals: ethylene, propylene, benzene from crude oil refining

■ Printing Technology

- Offset printing: ink transfers plate → blanket → paper (most common)
- Gravure printing: ink in recessed cells on cylinder → paper
- Screen printing: ink through mesh stencil — used for fabric/posters
- Digital/inkjet printing: direct from digital file, no plate required
- Letterpress: raised surface inked and pressed to paper (oldest)

d-4 CERAMIC ENGINEERING (2 Q | 2 M)

■ Classification of Pottery

- Earthenware: fired at low temp (~1000°C), porous, glazed — flower pots, tiles
- Stoneware: fired at ~1200°C, less porous, tough — drainage pipes
- Porcelain/China: fired at high temp ~1400°C, non-porous, translucent — tableware
- Bone china: porcelain + calcined bone ash — finest quality

■ Ceramic Materials & Properties

- Raw materials: clay (kaolinite), feldspar (flux), silica (quartz)
- Properties: high hardness, brittleness, high melting point, chemical resistance
- Production: Batching → Mixing → Forming → Drying → Glazing → Firing (sintering)
- Technical ceramics: alumina, silicon carbide — used in electronics, cutting tools

d-5 FOOD TECHNOLOGY (2 Q | 2 M)

■ Vitamins & Nutrition

- Fat-soluble vitamins: A (night blindness), D (rickets), E (antioxidant), K (clotting)
- Water-soluble: B-complex (B1-B12), C (scurvy) — not stored in body
- Cereals: rice, wheat, maize — provide carbohydrates, B vitamins
- Pulses (legumes): dal, chickpea, soybean — high protein content

■ Milk & Food Preservation

- Pasteurization: HTST (72°C/15s) or LTLT (63°C/30min) — kills pathogens
- Homogenization: fat globules broken for uniform consistency
- Milk products: cheese, butter, ghee, curd, paneer
- Food preservation methods: canning (sterilization), freezing, dehydration, pickling, smoking
- Food additives: preservatives (salt, sugar, vinegar), antioxidants, emulsifiers

d-6 OTHER ENGINEERING COURSES (4 Q | 4 M)

■ (i) Agriculture Engineering

- Farm power sources: draught animals, tractors, electric motors
- Tillage equipment: mouldboard plough, disc plough, cultivator, harrow
- Seeding: seed drill (line sowing), transplanter, broadcaster
- Harvesting: reaper, combine harvester (cuts + threshes + cleans)
- Irrigation equipment: surface, sprinkler system, drip/trickle irrigation

■ (ii) Architecture

- Ancient Indian: Indus Valley (brick cities: Harappa, Mohenjo-daro)
- Classical Hindu temples: Nagara (North — shikhara) vs Dravidian (South — gopuram)
- Mughal architecture: Red Fort, Taj Mahal, Humayun's Tomb — Persian influence
- Colonial (British): Victoria Memorial, India Gate — Indo-Saracenic style
- Building typologies: residential, commercial, institutional, industrial, recreational
- Architectural elements: arch, dome, vault, column, beam, lintel

■ (iii) Fashion Design & Technology

- History: ancient (draped clothing) → Renaissance → Industrial revolution → Modern
- Indian fashion eras: Vedic → Mughal → Colonial → Contemporary fusion
- Garment manufacturing machines: lockstitch sewing machine, overlock/serger
- Specialized machines: buttonhole, button-sewing, blind stitch, flatlock
- Basic tools: tailor's chalk, dress form/mannequin, pattern paper, notcher
- Fashion technology: CAD/CAM for pattern making (Gerber, Lectra systems)
- Fabric cutting: manual (scissors, rotary cutter) & automated (CNC cutter)